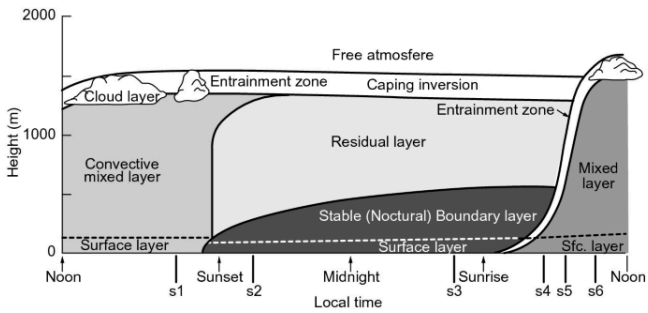


STRUCTURE OF THE ATMO BOUNDARY LAYER:

Boundary layer (BL): Part of the troposphere that is directly influenced by the presence of the earth's surface and responds to surface forcings with a timescale of about an hour of less.
Forcings like: frictional drag , evaporation and transpiration, heat transfer, pollutant emission, and terrain induced flow modifications. (Stull, 1950)

BL Thickness: varies in time space ranging from hundreds of meters to a few kilometers. There is a **diurnal variation** of temperature near the Earth's surface (depending on altitude) which plays a ket role in the characteristics of the boundary layer (BL). (Stull, 1950)
Over land surfaces in high pressure regions the BL has three defined structures that evolve during the diurnal cycle. These major components are the, **mixed layer**, **residual layer**, and the **stable boundary layer**. (Stull, 1950)



Surface layer: The region at the bottom of the boundary layer where turbulent fluxes and stress vary by less than 10% of their magnitude. The bottom 10% of the boundary layer (BL) is the surface layer regardless if it's in the mixing layer or the stable boundary layer. (Stull, 1950)

Mixed layer: Also known as the conductive boundary layer, the turbulence that happens within this mixing layer is **convectively driven** (heat rises from the ground, cooler temperatures sink towards the ground, another way of saying this is that it is driven by buoyancy), although a nearly well-mixed layer can form in regions with strong winds. Even if convection is the dominate source within the mixing layer (ML), there is usually **wind sheer** at the top of the ML that contributes to the turbulence within the layer.

*As mentioned above there is **convectively driven** turbulence, but there is also **mechanically driven** turbulence.

- EX of Convectively driven:** thermals or warm air rising
- EX of Mechanically driven:** wind shear and already existing turbulence

- Convectively driven > mechanically driven = **free convection**
Mainly occurs over land on clear sunny days with light calm winds.
Has a diurnal cycle, strong solar heating at the surface creates turbulence and depth within the mixing layer.
- Convectively driven < mechanically driven = **forced convection**
Mainly occurs on overcast days with strong winds

There is a stable layer at the top of the ML that acts like a "lid" to the rising thermals (conduction), thus restraining the domain of the turbulence. This layer is called the **entrainment zone**, because entrainment into the ML occurs there. (Stull, 1950)

*This entrainment zone controls the following processes of the boundary layer: The growth rate of the mixing layer, the turbulence budget, pollutant dilution, transfer of momentum downward, and transport of warm/dry air aloft.

*For winds within the ML, the middle portion has nearly constant speed and direction where it decreases towards zero near the ground. Resulting in a **logarithmic** wind profile with height in the surface layer. (Stull,1950)

Power Law Wind Profile

$$V_z = V_h \left(\frac{z}{h} \right)^p$$

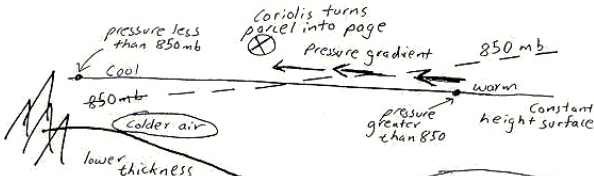
- Z= height at which the wind speed V is to be estimated
- h= height at which V_h was measured
- V=estimated wind speed
- p=constant ~ 1/7
- Note: P values of .125 for winds less than 50 kts and 0.08 for gusts and winds of at least 50 kts. However, this resulted in circumstances where the steady speeds exceeded the gusts. Therefore, a compromise of P = 0.10 is recommended for converting these wind speeds and gusts to other heights up to 100 m.

Residual layer: As the sunsets and the and thermal convection decays, so does the turbulence in the mixing layer start to decay. This results in the **Residual layer** because its initial mean state variables and concentration variables are the same as those recently-decayed mix layer. Meaning that the RL is neutrally stratified resulting in turbulence that is nearly equal in all directions and intensities. (Stull, 1950) I.e. calming conditions (?)

Stable Nocturnal layer: Also known as the **Stable Boundary Layer**, as the night progresses the bottom portion of the residual layer is transformed by its contact with the ground into a stable boundary layer. (Stull, 1950)

Main characteristics of the SNL: stable air with weaker, sporadic turbulence.

Yes, it's true that at night the winds near the ground are light. However they can become aloft and become accelerated into a **super geostrophic speeds** which is classified as a situation called a **low-level jet** or **nocturnal jet**. This nocturnal jet can enhance wind shear and generate turbulence. This turbulence happens in a short burst that could cause mixing through the stable boundary layer.



ATOMSPHERIC STABILITY:

Monin-Obukhov similarity theory (MOST): As discussed before, the surface layer of the boundary layer is where the fluxes within the layer vary by 10% of their magnitude with height. To a first order approximation, this layer is a **constant flux layer**.

With that, we can simplify the characteristics of the surface layer by utilizing at just one height – usually at the surface (Stull, 1950).

This similarity only works when the winds are NOT calm, and u* is NOT zero.

The relevant scales based on these fluxes and their typical orders of magnitude are listed here (Stull, 1950).

L	Order (1 m to 200 m)
z ₀	Order (1 mm to 1 m)
u _*	Order (0.05 to 0.3 m/s)
θ _* ^{SL}	Order (0.1 to 2.0 °C)
q _* ^{SL}	Order (0.1 to 5 g _{water} /kg _{air})

Where:

$$L = -\{ \overline{u'^2} + \overline{v'^2} \}^{3/4} / [k \cdot (g/\overline{\theta_v}) \cdot (\overline{w'\theta_v'})] = \text{Obukhov length}$$
$$L_L = -\{ \overline{u'^2} + \overline{v'^2} \}^{3/4} / [k \cdot (g/\overline{\theta_v}) \cdot (\overline{w'\theta_v'})] = \text{local Obukhov length}$$

u,v,w = eastward, northward, and upward wind component, respectively
θv = virtual potential temperature
g = gravity velocity
k= von Karman constant

$$z_0 = \text{aerodynamic roughness length}$$
$$u_* = \{ \overline{u'^2} + \overline{v'^2} \}^{1/4} = \text{friction velocity}$$
$$\theta_*^{SL} = -\overline{w'\theta_v'} / u_* = \text{surface-layer temperature scale}$$
$$q_*^{SL} = -\overline{w'q'} / u_* = \text{surface-layer humidity scale}$$

For the scales for pollutant concentration, they can be patterned after the humidity scale (Stull, 1950).

In plain terms, MOST is a relationship that describes the vertical behavior of nondimensionalized mean flow and turbulence properties within the atmospheric surface layer (the lowest 10% or the atmospheric boundary layer) as a function of the parameters listed above (AMS, 2026).

Richardson number (Ri) : Generally, it's a dimensionless ratio that describes the relative importance of convection and the turbulence generated by **mechanical sheer**. (Schnelle, 2003)

The basic understanding of the Richardson number (Ri) can be described as the following equation.

$$R_i = \frac{g}{\rho} \frac{\partial \rho}{\partial z} \bigg/ \left(\frac{\partial u}{\partial z} \right)^2 ,$$

Where:

- g = gravity
- (rho) = density
- z = height/depth
- u/z = the change of wind speed with height

This value can be used to determine if the flow of the atmosphere (when measured) is laminal/stable or turbulent/unstable.

The following Ri values can be used as a rule-of-thumb if the atmosphere is stable or unstable: (SKYbrary, 2026)

- Ri > 0.25 --> Stable; No vertical mixing, weak winds.
- 0 < Ri < 0.25 --> Stable, mechanical turbulence weakened by stable stratification
- Ri = 0 --> Neutral, Mechanical turbulence only
- 0.03 < Ri < 0 --> Unstable, Mechanical turbulence and convection
- Ri < -0.04 --> Unstable, convection predominate, weak winds, strong vertical motion, pollution concentrations and plumes rapidly spread vertically and horizontally.

However there are different variations of the Richardson number that also can relate to the stabilities within the atmospheric boundary layer.

Bulk Richardson Number (RB): an approximation to Ri formed by approximating the local gradients by a finite difference across the atmospheric layers. Used in the context of convective storm forecasting, the non-dimensional ratio of convective available potential energy (CAPE) to measure **vertical wind shear**, which can be used to characterized convective-storm types for various environments (AMS, 2026).

$$R_b = \frac{(g/T_v) \Delta \theta_v \Delta z}{(\Delta U)^2 + (\Delta V)^2},$$

Where:

- Tv = absolute virtual temperature
- θv = virtual potential temperature
- Δz = thickness across different layers
- ΔU, ΔV = changes in the horizontal wind components.

Flux Richardson Number (Rf): Another dimensionless number that is defined as the ratio of buoyancy term of the turbulent kinetic energy (TKE) budge and the negative shear terms from the same budget TKE formula. Used in the context to measure dynamic stability to capture the capability of the flow to remain turbulent by wind shear overpowering any static stability (AMS, 2026).

$$R_f = \frac{(g/T_v) \overline{w'\theta_v'}}{\overline{u'w'} (\partial \overline{U} / \partial z) + \overline{v'w'} (\partial \overline{V} / \partial z)},$$

Obukhov length: A parameter with dimension of length that, when used to scale the height above the ground z, yields a dimensionless stability parameter, z/L, that expresses the relative roles of shear and buoyancy in the production/consumption of turbulent kinetic energy (TKE). Which can be defined as: (AMS,2026)

$$L = \frac{-u_*^3 T_v}{kg Q_{r0}},$$

Where:

- k = Von Karman's constant
- u* = friction velocity (a measure of surface stress)
- Q_v0 = kinematical virtual temperature flux at the surface
- Tv is a reference virtual temperature
- g = gravitational acceleration
- g/Tv = a ratio that define the buoyancy parameter
- Tv = the surface air temperature

Sidenote: This NOT the same as the length parameter that is used in the Monin-Obukhov similarity theory!



THE NOCTURNAL LOW LEVEL JET

But, during the nonturbulent periods, the stable boundary layer becomes decoupled from the surface. (Stull,1950)

TURBULENCE BASICS:

Wind speed varies in an irregular pattern – a characteristic signature of turbulence. This quasi-randomness is what makes turbulence different from other motions like waves (Stull, 1950). Depending on the time of day and the local conditions, winds can statistically show that they have a running-average trend through the times of the day. The ability to find a statistically-stable mean value for winds suggest that turbulence in NOT completely random (Stull, 1950) and can be quantified through statistics.

Turbulent Kinetic Energy (TKE): This important value within the boundary layer can be used to determine the time line of the magnitude of the turbulence within the layer. One of the reason why TKE is important for understanding micrometeorology and wind science is that it's used as a measurement of turbulence. TKE is directly related to the elements of momentum heat and moisture transport throughout the boundary layer (Stull, 1950).

Whenever turbulence is created buoyant thermals and/or mechanical eddies, it can be suppressed by the stable lapse rate and dissipate into heat by the effect of molecular viscosity. (Stull,1950) By deriving an equation a budget for the TKE, we can calculate a production terms against the loss to determine if the boundary layer will become *more* turbulent or if the turbulence will *decay* within the boundary layer. (Stull,1950) The derived equation is as follows:

$$TKE = \frac{1}{2} (\sigma_u^2 + \sigma_v^2 + \sigma_w^2),$$

where σ_u^2 , σ_v^2 , and σ_w^2 represent the variance of the latitudinal (u), longitudinal (v), and vertical (w) velocities (m/s). Based on the values obtained with this equation, the atmosphere is neutral when $0.7 < TKE < 1.0$, stable when $TKE < 0.7$, and unstable when $TKE > 1.0$ (in m^2s^{-2}) (Perez Albornoz et al., 2022).

However the budget of the TKE is expressed with more than just the latitudinal, longitudinal and the vertical velocities within the boundary layer. By using the prognostic equation for the sum of velocity variances and dividing by two gives us the following TKE budget equation (Stull,1950):

$$\frac{\partial \bar{e}}{\partial t} + \bar{U}_j \frac{\partial \bar{e}}{\partial x_j} = + \delta_{ij} \frac{g}{\theta_v} (\overline{u_i' \theta_v'}) - \overline{u_i' u_j'} \frac{\partial \bar{U}_i}{\partial x_j} - \frac{\partial (\overline{u_j' e})}{\partial x_j} - \frac{1}{\bar{p}} \frac{\partial (\overline{u_j' p'})}{\partial x_j} - \epsilon \quad (5.1a)$$

I	II	III	IV	V	VI	VII
Term I	Term II	Term III	Term IV	Term V	Term VI	Term VII
represents local <i>storage</i> or tendency of TKE. describes the <i>advection</i> of TKE by the mean wind. is the <i>buoyant production or consumption term</i>. It is a production or loss term depending on whether the heat flux $\overline{u_i' \theta_v'}$ is positive (during daytime over land) or negative (at night over land). is a <i>mechanical or shear production/loss term</i>. The momentum flux $\overline{u_i' u_j'}$ is usually of opposite sign from the mean wind shear, because the momentum of the wind is usually lost downward to the ground. Thus, Term IV results in a positive contribution to TKE when multiplied by a negative sign. represents the <i>turbulent transport</i> of TKE. It describes how TKE is moved around by the turbulent eddies u_j'. is a <i>pressure correlation term</i> that describes how TKE is redistributed by pressure perturbations. It is often associated with oscillations in the air (<i>buoyancy or gravity waves</i>). represents the viscous <i>dissipation</i> of TKE; i.e., the conversion of TKE into heat.						

If we choose a coordinate system aligned with the mean wind, assume horizontal homogeneity, and neglect subsidence, then a special form of the TKE budget equation can be written

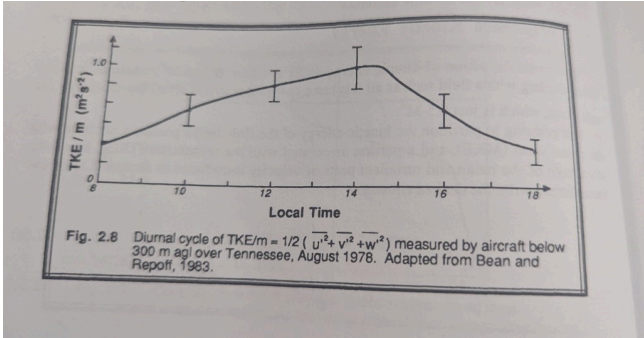
$$\frac{\partial \bar{e}}{\partial t} = \frac{g}{\theta_v} (\overline{w' \theta_v'}) - \overline{u' w'} \frac{\partial \bar{U}}{\partial z} - \frac{\partial (\overline{w' e})}{\partial z} - \frac{1}{\bar{p}} \frac{\partial (\overline{w' p'})}{\partial z} - \epsilon \quad (5.1b)$$

I	III	IV	V	VI	VII
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(Stull, 1950)

From the fully derived equations above, TERM VII shows that turbulence is **dissipative**, meaning that there is always a loss value to the TKE whenever TKE is not zero. Meaning that the turbulence will tend to decay and disappear over time, unless it can be generated locally or transported in by mean, turbulent, or pressure process. Meaning that the value of TKE is NEVER conserved. (Stull,1950)

The usage of the derived TKE equation allows us to plot the conductive conditions within the mixing layer throughout the time of day. (Stull, 1950) As show in the plot below, during the daytime, the buoyancy allows the air parcels to accelerate in the middle of the mixing layer. (Stull,1950)



Eddies: A function and motion of turbulent transport. Can be visualized as the gustiness superimposed onto the mean wind as irregular swirls. Mainly, turbulence consist of many different size eddies that feed into each other. The relative strength of theses different size eddies can be defined as **the turbulence spectrum** (Stull,1950).

Mathematically, the turbulence spectrum can be calculated to how the different size eddies feed into the total turbulent kinetic energy (TKE). With this, as a functions of Eddie strength over time and the frequency of wind-speed variation, we can derive that small eddies have a shorter time period than large eddies. This conclusion is apart of the **Taylor's hypothesis** (Stull,1950).

Taylor's hypothesis: In1938, G. I. Taylor suggest that turbulence might be considered to be *frozen* as it advects past a meteorological instrument sensor. Meaning, that we can use the measured wind speed to translated the turbulence measurement as a function of time to the corresponding measurement in space. In reality though, turbulence is never really frozen, the simplification is useful to make calculations for cases where the turbulent eddies

WIND SHEAR AND VEER:

Wind shear: Frictional drag on the air flowing over the ground causes wind shear to develop, which causes the air flow to become turbulent. (Stull, 1950). Or simply, the change in wind speed with height, with irregular topography, it can lead to significant variations in wind **speeds** at different altitudes. (Pardo, 2025)

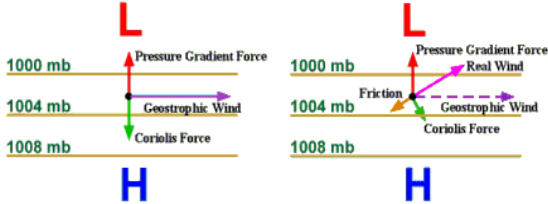
Wind veer: Similar to *wind shear*, it's the change of wind **direction** with height. (Pardo, 2025)

Understanding both wind shear and veer are important for understanding and predicting wind behaviors. As mentioned above th surrounding terrain can have an impact on the variances of theses behaviors. For example, valley or mountainous area can lead to [wind] shear and/veer to become highly variable (Pardo, 2025).

Vertical wind change in speed/direction:

Their causes:

Surface friction: Has a critical role on the speed and direction of surface winds. Depending how rough the ground is, the air slows down as it moves over the ground, wind speeds are less than would be expected from the pressure gradient on the weather map and the direction is changed so that the wind blows across the isobars into a center of low pressure and out o a center of high pressure. (weather.gov, 2026)



The surface friction effect usually does not extend more than a couple thousand feet into the air.

For example: at 3000 feet above the ground, the wind blows parallel to the isobars with a speed proportional to the pressure gradient.

Accounting for the effects of surface friction, winds, locally, do not always show the speed and direction that would be expected from the isobars on the surface weather map. The differences are mainly due to the geographical features such as hills, mountains, and large bodies of water.

For mountainous regions, the effect of the terrain that would cause variations to the wind behaviors (speed and direction) to extend no higher than about 2000 feet above the ground (weather.gov, 2026)

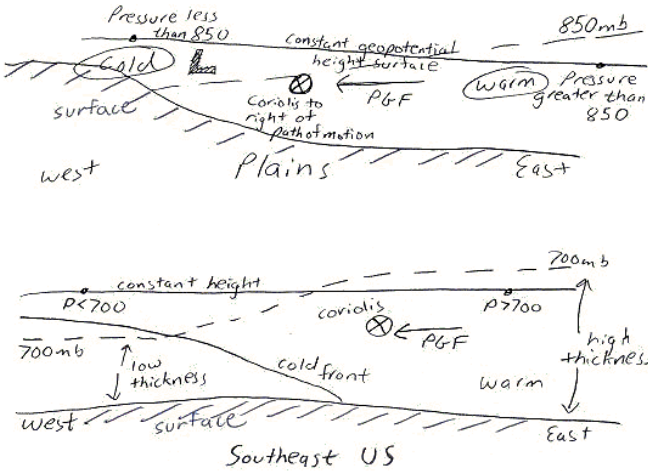
Thermal stratification: Induced by solar radiation, it is the uneven temperature distribution across the atmospheric profile. Depending on the variousness of this thermal distribution, it could lead to stability or instability within the atmosphere. The stability of this thermal stratification can lead to how wind behaviors (speed and direction) are dispersed within the atmospheric profile (Yang et al., 2024).

Low-level jets: A fast moving ribbon-like mass of air in the lower levels of the atmosphere with speeds that range from 25 to over 70 knots.

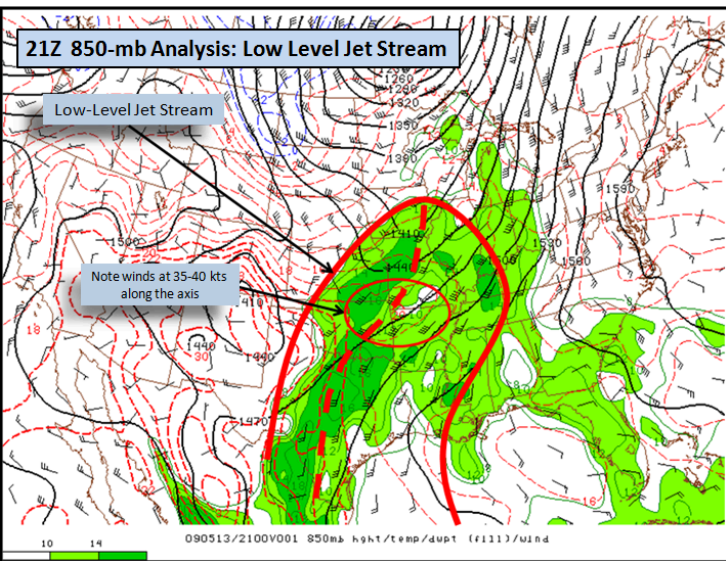
There are two main types of low-level jets. The **nocturnal low level jet** and the **mid-latitude cyclone induced low level jet**. (weather.gov, 2026)

Nocturnal low level jet: (See Stable Nocturnal Layer)

Mid- latitude cyclone induced low level jet: Generally stronger than the nocturnal low level jet, generally due to a combination of the nocturnal jet and thermal winds forcing towards a low pressure system. Mainly cause by large temperature gradient expanding from behind a cold front to a warm sector of a cyclone. (weather.gov, 2026)



THE CYCLONE INDUCED LOW LEVEL JET



TERRIAN AND SURFACE-ROUGHNESS EFFECT ON NEAR SURFACE FLOWS

Land and Sea Breezes: Land and sea breezes are caused by the differences in temperature over land and water. The sea breeze occurs during the day when the land area heats more rapidly than the water surface. This results in the pressure over the land being lower than that over the water. The pressure gradient is often strong enough for a wind to blow from the water to the land. (weather.gov.2026)



evolve with a timescale longer than the time it takes the eddy to be advected past a MET sensor (Powell and Elderkin, 1974).

Reynolds Decomposition: (** struggled to understand this from the Stull,1950 book so used CoPilot)

A mathematical practice that can separate chaotic atmospheric variable (i.e., wind speed, temperature, and humidity) into two computable parts, a **mean component** (the average over a period of time), and a **fluctuating component** (the fluctuation of the variable caused by turbulence).

For example for horizontal winds of u it can be broken down into:

$$u = U' + u'$$

Where:

U' is the mean wind

u' is the fluctuating wind from turbulence.

In discussion of the atmospheric boundary layer, the Reynold decomposition splits the instantaneous velocities in the mean part of the boundary layer profile (depending on the altitude) and the turbulent velocity fluctuations over the temporal averages to make the calculations of a statistical mean of irregular atmospheric variables possible. (Eric Vantroyen, 2020: [Turbulence/ Intensity. Reynolds decomposition - Wind resources for renewable energies](#))

How this is helpful for micro-meteorology and wind science is that it can be used from Eddy covariance measurements from measurements from MET towers.

For example:

Sensible heat flux = (vertical wind fluctuation) x (temperature fluctuation), averaged over time

This method works because turbulence transports properties through correlated gusts and eddies.

The Reynolds decomposition allows for turbulence to be parameterized in weather and climate models which allows the models to show closure schemes that relate turbulent fluxes to mean gradients, an estimation of mixing coefficients, and a representation of surface-atmosphere exchanges.

Turbulence Intensity: Defined as a dimensionless ratio of the standard deviation of wind speed to the mean wind speed, typically measure over a period of 10 minutes. It reflects the instantaneous, random deviations from the mean wind speed and varies with factors such as surface roughness and atmospheric stability. (Saeidi et al., 2011)

The calculation of Turbulence Intensity (TI) can be defined as:

$$\text{TI} = \frac{\sigma_U}{\bar{U}}$$

where $(\omega)_U$ and $\bar{U}$ represent the standard deviation and mean wind speed, respectively (Saeidi et al., 2011).

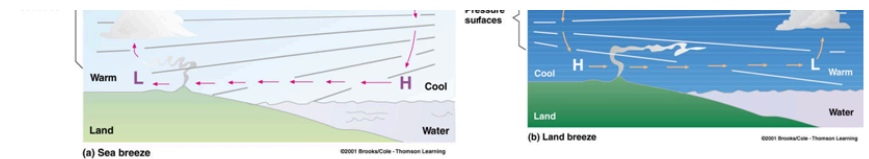
Turbulence intensity changes with the mean wind speed, with surface roughness, with atmospheric stability and with the topographic features. (Saeidi et al., 2011)

However, for this value to support Taylor's hypothesis, $TI < 0.5$.

DIURNAL CYCLES ON THE ABL AND ITS IMPACT:

Diurnal Atmospheric Boundary Layer: (see Boundary Layer, TKE, low-level jets); The entire atmosphere gets the majority of its energy from solar radiation (shortwave energy). Which is primarily absorbed by the Earth's surface and then transmitted (longwave energy) to the rest of the atmosphere through the boundary layer process. (Stull, 1950)

About 90% of the net radiation is absorbed by the oceans causing evaporation (water vapor). The latent heat stored in this water vapor accounts for 80% of the fuel that drives the movement within the atmosphere. (Stull,1950)

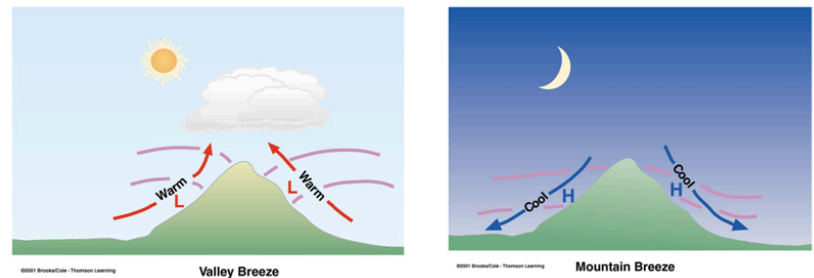


Mainly happens in local regions that are located in a narrow are along a coast.

The breeze blows towards the land at night when the surface is cooler. Then the breeze blows towards the sea during the day towards the warm, low-pressure area over the water. (weather.gov, 2026)

Mountain Winds: Hills and valleys substantially distort the airflow associated with the prevailing pressure system and the pressure gradient.

Strong up and down drafts and **eddies** develop as the air flows up over hills and down into valleys. Wind direction changes as the air flows around hills. Sometimes lines of hills and mountain ranges will act as a barrier, holding back the wind and deflecting it so that it flows parallel to the range. If there is a pass in the mountain range, the wind will rush through this pass as through a tunnel with considerable speed. The airflow can be expected to remain **turbulent and erratic** for some distance as it flows out of the hilly area and into the flatter countryside. (weather.gov, 2026)



Daytime heating and nighttime cooling of the hilly slopes lead to day to night variations in the airflow.

At night, the sides of the hills cool by radiation. The air in contact with them becomes cooler and therefore denser and it blows down the slope into the valley. This is a **katabatic wind** (sometimes also called a mountain breeze).

If the slopes are covered with ice and snow, the katabatic wind will blow, not only at night, but also during the day, carrying the cold dense air into the warmer valleys.

The slopes of hills not covered by snow will be warmed during the day. The air in contact with them becomes warmer and less dense and, therefore, flows up the slope. This is an ***anabatic wind (or valley breeze)***. (weather.gov,2026)

In local mountainous area the distortion and **distribution of airflow** is more severe. Rocky surfaces, high ridges, sheer cliffs, steep valleys, all combine produce unpredictable flow patterns and turbulence. (weather.gov,2026)



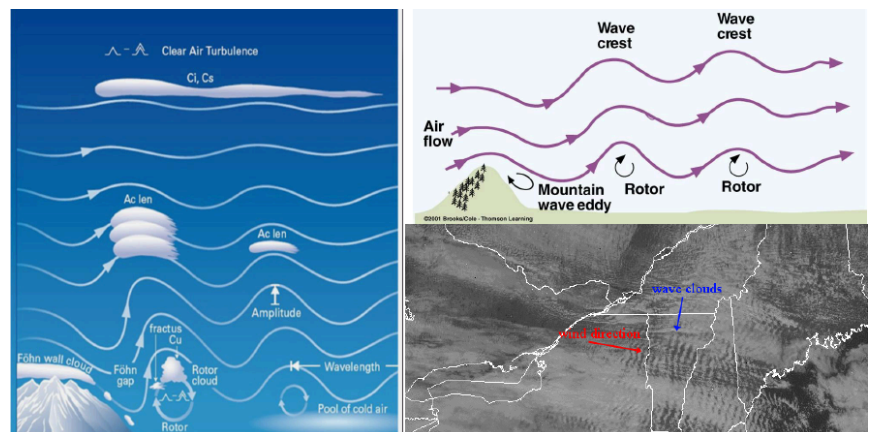
The Mountain Wave: Air flowing across a mountain range usually rises relatively smoothly up the slope of the range, but, once over the top, it pours down the other side with considerable force, bouncing up and down, creating **eddies and turbulence** and also creating powerful vertical waves that may extend for great distances downwind of the mountain range. This phenomenon is known as a mountain wave. Note the up and down drafts and the rotating eddies formed downstream.

If the air mass has a high moisture content, clouds of very distinctive appearance will develop.

Cap Cloud. Orographic lift causes a cloud to form along the top of the ridge. The wind carries this cloud down along the leeward slope where it dissipates through adiabatic heating. The base of this cloud lies near or below the peaks of the ridge; the top may reach a few thousand feet above the peaks.

Lenticular (Lens Shaped) Clouds form in the wave crests aloft and lie in bands that may extend to well above 40,000 feet.

Rotor Clouds form in the rolling eddies downstream. They resemble a long line of stratocumulus clouds, the bases of which lie below the mountain peaks and the tops of which may reach to a considerable height above the peaks. Occasionally these clouds develop into thunderstorms. (weather.gov, 2026)



The clouds, being very distinctive, can be seen from a great distance and provide a visible warning of the mountain wave condition. Unfortunately, sometimes they are embedded in other cloud systems and are hidden from sight. Sometimes the air mass is very dry and the clouds do not develop.

The severity of the mountain wave and the height to which the disturbance of the air is affected is dependent on the strength of the wind, its angle to the range and the stability or instability of the air. The most severe mountain wave conditions are created in strong airflows that are blowing at right angles to the range and in stable air. A jet stream blowing nearly perpendicular to the mountain range increases the severity of the wave condition.

The mountain wave phenomenon is not limited only to high mountain ranges, such as the Rockies, but is also present to a lesser degree in smaller mountain systems and even in lines of small hills. (weather.gov, 2026)